Customer Shopping Behavior Analysis

Bilingual Portfolio Report | Reporte bilingüe para portafolio

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A. Executive Summary / Resumen Ejecutivo

This project examines 3,900 retail transactions to identify patterns in spending, customer loyalty, product performance, discount behavior, and subscription engagement.
Este proyecto examina 3,900 transacciones retail para identificar patrones de gasto, lealtad del cliente, desempeño de productos, comportamiento ante descuentos y suscripciones.

The workflow combines Python for data cleaning and feature engineering, PostgreSQL for business analysis, and Power BI for dashboard storytelling.
El flujo de trabajo combina Python para limpieza y transformación de datos, PostgreSQL para análisis de negocio y Power BI para visualización ejecutiva.

Portfolio Value / Valor para Portafolio

- Shows end-to-end analytics skills: data preparation, SQL analysis, BI storytelling, and business recommendations.
  Demuestra habilidades analytics de punta a punta: preparación de datos, análisis SQL, storytelling en BI y recomendaciones de negocio.

Quick Highlights / Hallazgos Clave

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Dataset size / Tamaño	3,900 purchases	Large enough to observe customer patterns across categories and segments.
Missing values / Valores faltantes	37 review ratings	Required imputation before modeling and analysis.
Subscribers / Suscriptores	1,053	Helpful for loyalty and recurring-value analysis.
Non-subscribers / No suscriptores	2,847	Main volume of customers and revenue base.

1. Project Overview / Descripción del Proyecto

The project analyzes customer shopping behavior using transactional data across multiple categories. The main goal is to uncover spending patterns, customer segments, product preferences, and subscription behavior that can guide strategic decisions.
El proyecto analiza el comportamiento de compra del cliente usando datos transaccionales en múltiples categorías. El objetivo principal es descubrir patrones de gasto, segmentos de clientes, preferencias de producto y comportamiento de suscripción que apoyen decisiones estratégicas.

Business Question / Pregunta de Negocio

How can the company use consumer shopping data to identify trends, improve customer engagement, and optimize marketing and product strategies?
¿Cómo puede la empresa usar datos de compra del consumidor para identificar tendencias, mejorar la interacción con clientes y optimizar estrategias de marketing y producto?

2. Dataset Summary / Resumen del Dataset

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Rows / Filas	3,900	Represents the number of purchases available for analysis.
Columns / Columnas	18	Includes demographic, transaction, and behavior variables.
Core attributes / Atributos clave	Age, Gender, Location, Category, Review Rating	Useful for segmentation, product analysis, and customer profiling.
Behavior fields / Variables de comportamiento	Discount Applied, Previous Purchases, Frequency, Shipping Type	Helpful to understand repeat buying and purchase drivers.

Main Feature Groups / Grupos Principales de Variables

- Customer demographics: Age, Gender, Location, Subscription Status.
  Datos demográficos del cliente: edad, género, ubicación y estado de suscripción.
- Purchase details: Item Purchased, Category, Purchase Amount, Season, Size, and Color.
  Detalles de compra: artículo comprado, categoría, monto, temporada, talla y color.
- Shopping behavior: Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, and Shipping Type.
  Comportamiento de compra: descuento aplicado, código promocional, compras previas, frecuencia de compra, calificación y tipo de envío.

3. Data Preparation and Exploratory Analysis (Python) / Preparación de Datos y Análisis Exploratorio (Python)

Python was used to prepare the raw data before running structured SQL analysis. This stage focused on data quality, naming consistency, and feature creation.
Python se utilizó para preparar los datos en bruto antes del análisis estructurado en SQL. Esta etapa se enfocó en calidad de datos, consistencia de nombres y creación de variables.

- Data loading with pandas.
  Carga de datos con pandas.

• Initial exploration with df.info() and df.describe().
Exploración inicial con df.info() y df.describe().

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN

• Missing value treatment: 37 missing values in Review Rating were imputed using the median rating by product category.
Tratamiento de faltantes: los 37 valores faltantes en Review Rating se imputaron usando la mediana por categoría de producto.

• Column standardization: fields were renamed to snake_case for readability and documentation.
Estandarización de columnas: los campos se renombraron a snake_case para mejorar legibilidad y documentación.

• Feature engineering: age_group and purchase_frequency_days were created to support segmentation analysis.
Ingeniería de variables: se crearon age_group y purchase_frequency_days para apoyar el análisis de segmentación.

• Consistency check: promo_code_used was removed after verifying redundancy with discount_applied.
Validación de consistencia: promo_code_used se eliminó después de verificar redundancia con discount_applied.

• Database integration: the cleaned dataframe was loaded into PostgreSQL for SQL analysis.
Integración con base de datos: el dataframe limpio se cargó en PostgreSQL para análisis SQL.

4. Business Analysis Using SQL / Análisis de Negocio Usando SQL

PostgreSQL was used to answer business questions related to revenue, customer segments, product performance, shipping behavior, subscriptions, discounts, and age-based revenue contribution.
PostgreSQL se utilizó para responder preguntas de negocio relacionadas con ingresos, segmentos de clientes, desempeño de productos, comportamiento de envío, suscripciones, descuentos y contribución de ingresos por edad.

4.1 Revenue by Gender / 4.1 Ingresos por Género

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Female revenue / Ingresos mujeres	75,191	Female customers generated a meaningful share of revenue.

Male revenue / Ingresos hombres	157,890	Male customers generated higher total revenue in this dataset.

This comparison helps identify where marketing and assortment strategies may need to be differentiated by gender.
Esta comparación ayuda a identificar dónde las estrategias de marketing y surtido pueden diferenciarse por género.

	gender text	revenue numeric
1	Female	75191
2	Male	157890

4.2 High-Spending Discount Users / 4.2 Clientes con Alto Gasto y Descuento

The analysis identified customers who used discounts but still spent above the average purchase amount. This suggests that discounts can attract high-value buyers, not only bargain shoppers.
El análisis identificó clientes que usaron descuentos pero aun así gastaron por encima del promedio. Esto sugiere que los descuentos pueden atraer compradores de alto valor, no solo clientes sensibles al precio.

	customer_id bigint	purchase_amount bigint
1	2	64
2	3	73
3	4	90
4	7	85
5	9	97
6	12	68
7	13	72
8	16	81
9	20	90
10	22	62

4.3 Top 5 Products by Rating / 4.3 Top 5 Productos por Calificación

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Gloves / Guantes	3.86	Highest average product rating.

Sandals / Sandalias	3.84	Strong perceived product quality.
Boots / Botas	3.82	Good candidate for promotional positioning.
Hat / Sombrero	3.80	High-rated product with broad appeal.
Skirt / Falda	3.78	Useful for merchandising and featured campaigns.

4.4 Shipping Type Comparison / 4.4 Comparación por Tipo de Envío

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Standard / Estándar	58.46	Lower average purchase amount.
Express / Exprés	60.48	Slightly higher average order value.

Customers using express shipping show a marginally higher average spend, which may indicate stronger purchase intent or time-sensitive buying behavior.
Los clientes que usan envío exprés muestran un gasto promedio ligeramente mayor, lo cual puede indicar mayor intención de compra o comportamiento más urgente.

4.5 Subscribers vs. Non-Subscribers / 4.5 Suscriptores vs. No Suscriptores

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Subscribers / Suscriptores	1,053 customers | Avg 59.49 | Revenue 62,645	Subscribers represent a smaller but valuable group.
Non-subscribers / No suscriptores	2,847 customers | Avg 59.87 | Revenue 170,436	Non-subscribers drive most of total revenue by volume.

4.6 Discount-Dependent Products / 4.6 Productos Dependientes del Descuento

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia

Hat / Sombrero	50.00%	Most discount-dependent item in the sample.
Sneakers / Tenis	49.66%	High reliance on discounted purchases.
Coat / Abrigo	49.07%	Discounts strongly influence conversion.
Sweater / Suéter	48.17%	Pricing strategy matters for this item.
Pants / Pantalones	47.37%	Promotion-sensitive category.

These products may need careful margin management, stronger pricing strategy, or targeted promotions instead of blanket discounting.
Estos productos pueden requerir control de márgenes, una mejor estrategia de precios o promociones dirigidas en lugar de descuentos generalizados.

4.7 Customer Segmentation / 4.7 Segmentación de Clientes

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Loyal / Leales	3,116	Largest segment; priority for retention and upsell.
Returning / Recurrentes	701	Strong conversion opportunity into loyalty programs.
New / Nuevos	83	Smallest segment; acquisition funnel can be improved.

4.8 Top 3 Products per Category / 4.8 Top 3 Productos por Categoría

- Accessories: Jewelry, Sunglasses, Belt.
  Accesorios: Jewelry, Sunglasses y Belt.
- Clothing: Blouse, Pants, Shirt.
  Ropa: Blouse, Pants y Shirt.
- Footwear: Sandals, Shoes, Sneakers.
  Calzado: Sandals, Shoes y Sneakers.
- Outerwear: Jacket, Coat (visible in the original table).
  Ropa exterior: Jacket y Coat (visibles en la tabla original).

4.9 Repeat Buyers and Subscriptions / 4.9 Compradores Recurrentes y Suscripciones

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Repeat buyers without subscription / Recurrentes sin suscripción	2,518	Most repeat buyers are not subscribed yet.
Repeat buyers with subscription / Recurrentes con suscripción	958	Clear opportunity to expand loyalty conversion.

4.10 Revenue by Age Group / 4.10 Ingresos por Grupo de Edad

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Young Adult / Adulto joven	62,143	Highest total revenue contribution.
Middle-aged / Mediana edad	59,197	Second-highest revenue segment.
Adult / Adulto	55,978	Consistent purchasing power.
Senior / Senior	55,763	Comparable contribution to adult customers.

5. Power BI Dashboard / Dashboard en Power BI

The final stage of the project was an interactive dashboard built in Power BI to communicate insights visually and support faster decision-making.
La etapa final del proyecto fue un dashboard interactivo en Power BI para comunicar hallazgos de manera visual y apoyar decisiones más rápidas.

Insight / Hallazgo	Value / Valor	Why it matters / Relevancia
Number of customers / Número de clientes	3.9K	Confirms the scale of the analysis.
Average purchase amount / Ticket promedio	$59.76	Useful KPI for pricing and sales strategy.
Average review rating / Calificación promedio	3.75	General benchmark of customer satisfaction.

Subscription split / Distribución de suscripción	Yes 27% | No 73%	Important for retention and loyalty planning.

- The dashboard also visualizes revenue by category, sales by category, revenue by age group, and sales by age group.

El dashboard también visualiza ingresos por categoría, ventas por categoría, ingresos por grupo de edad y ventas por grupo de edad.

6. Business Recommendations / Recomendaciones de Negocio

- Boost subscriptions by promoting exclusive benefits and higher-value retention offers.

Impulsar las suscripciones mediante beneficios exclusivos y ofertas de retención de mayor valor.

- Create or strengthen loyalty programs to convert returning buyers into loyal customers.

Crear o fortalecer programas de lealtad para convertir compradores recurrentes en clientes leales.

- Review discount policy to balance conversion improvement with margin control.

Revisar la política de descuentos para equilibrar incremento en conversión con control de márgenes.

- Highlight top-rated and best-selling products in campaigns and product positioning.

Destacar productos mejor calificados y más vendidos en campañas y posicionamiento.

- Target high-revenue age groups and customers who prefer express shipping.

Enfocar esfuerzos en grupos de edad con mayor ingreso y clientes que prefieren envío exprés.

Recruiter Framing / Cómo Presentarlo a Reclutadores

- This project shows that I can move from raw data to business-facing recommendations.

Este proyecto demuestra que puedo avanzar de datos en bruto a recomendaciones orientadas al negocio.

- It also highlights practical use of Python, SQL, and Power BI within a single analytics workflow.

También resalta el uso práctico de Python, SQL y Power BI dentro de un mismo flujo analytics.